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Research review paper

A little breath of fresh air into an anaerobic system: How microaeration facilitates anaerobic digestion process

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ABSTRACT

Exposure of a small amount of oxygen/air (microaeration) has been reported to benefit the anaerobic digestion (AD) process in enhancing hydrolysis, improving methane yield, stabilizing the process and scavenging hydrogen sulfide among others. The underlying mechanism of enhancing AD process via microaeration is the augmentation of activity and diversity of the microbial consortia that promotes syntrophic interactions among different microbial groups, thereby creating a more stable process. To design and implement a microaeration-based AD process, fundamental insights about the mechanism of the AD system at process, microbial and molecular levels must be fully explored. This review critically examines microaeration-based AD processes through our recent understandings of the effect of oxygen on microbial community structure, enzymatic, energetic, physiological, and biochemical aspects of the microbial-mediated process. Syntrophic interactions between hydrolytic, fermentative, sulfate reducing, syntrophic bacteria and methanogens under microaerobic conditions are examined to reveal putative mechanism and factors that need to be considered when implementing microaeration in AD process. Further studies are needed to better understand the microbial pathways and bioenergetics of the microaerobic AD process by adopting advanced molecular techniques such as metagenomics, transcriptomics, and proteomics.

1. Introduction

Anaerobic digestion (AD), a mature technology originally employed for waste stabilization and waste (water) remediation, has gained significant attention in recent years for producing bioenergy from diverse feedstocks. Besides the source of renewable energy resources, AD process has also been used for production of diverse biobased products such as biopolymers, valuable biochemicals via carboxylic acids platform and organic fertilizer among other products using biorefinery platform (Sawatdeenarunat et al., 2016; Surendra et al., 2015). Moreover, AD process can also curtail greenhouse gases (GHGs) emissions by replacing part of fossil fuels, minimizing deforestation for fire woods in developing countries, and diverting food wastes, and organic fraction of municipal solid waste from landfills. These inherent merits make AD process highly desirable technology for remediating organic wastes, with concomitant production of renewable energy and value-added products.

Despite the aforementioned merits, the AD process has several drawbacks such as low hydrolysis rate, unstable at high organic loading rate (OLR), and release of toxic/corrosive hydrogen sulfide; however, all these issues could be overcome through micro-aeration. AD process could suffers from volatile fatty acids (VFAs) built-up due to involvement of highly sensitive microbial communities and delicate balance among interrelated biochemical processes, especially at high organic loading rates (Nguyen et al., 2015). The anaerobic degradation of organic matters follow a series of interdependent biochemical steps (i.e., hydrolysis, acidogenesis, syntrophic acetogenesis, and methanogenesis) carried out by diverse microbial groups, in which metabolic products excreted by one group of microorganisms are vital substrates for another group of microorganisms (Khanal and Li, 2017). For effective performance and stable operation of an AD system, all these interdependent stages should be kept kinetically and energetically at equilibrium by maintaining the syntrophic activities among the microbiomes. For example, in case of AD of highly recalcitrant feedstocks such as lignocellulosic biomass (e.g., agricultural waste/residues, forest residues and energy crops), enhancing hydrolysis is critical to solubilize these complex substrates into simple organic substrates for further downstream conversion (Mathews et al., 2015; Shrestha et al., 2017). On the other hand, AD of readily degradable substrates such as food wastes and food processing wastewater at high organic loading rate (OLR) is challenging due to the rapid production of fermentative products, especially VFAs. The accumulation of VFAs inhibits methanogens

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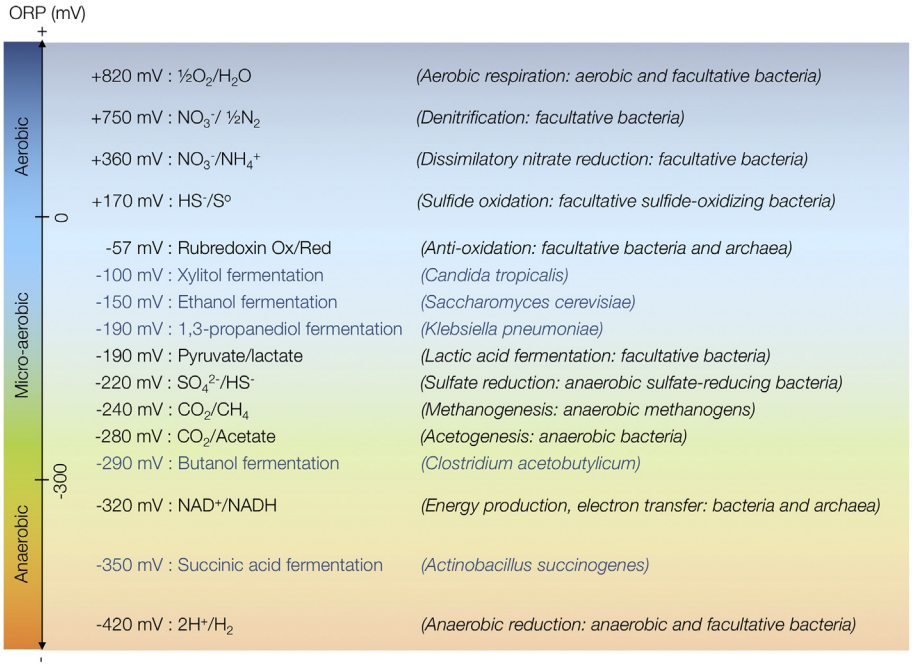


Fig. 1. ORP (mV with SHE) of redox couples and various microorganisms. The ORP values and positions of text are not to scale with axis. Values from Madigan et al. (2015) are in black text and values from Liu et al. (2013) are in blue text. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

that leads to complete failure of the AD process (Boe and Angelidaki, 2012; Nguyen et al., 2015). Additionally, during anaerobic treatment of sulfate-rich wastewater, hydrogen sulfide is generated via dissimilatory sulfate reduction, which not only inhibits methanogens but also causes corrosion of pipes and other metal parts, and needs to be removed from both aqueous and gaseous phases (Krayzelova et al., 2015). Fortunately, recent advances in microaeration technology have the potential to address above stated drawbacks of AD process, thereby opening-up a potential innovative solution in AD process control and stability.

Traditionally, air or oxygen exposure to AD system is avoided due to its negative effects on growth and activity of obligate anaerobes, especially methanogens. In recent years, however, numerous studies reported the beneficial effects on AD process when a limited amount of air or oxygen is injected, commonly termed as microaeration. The positive effects of microaeration include scavenging hydrogen sulfide, enhancing hydrolysis, and maintaining low VFAs concentration, thereby improving the overall stability of AD systems (Khanal and Huang, 2006; Lim et al., 2014; Ramos and Fdz-Polanco, 2013; Xu et al., 2014). The only existing full-scale application of microaeration system is biogas desulfurization, typically found in municipal wastewater treatment plants in central Europe, where compressed air is injected in sludge recirculation line of an anaerobic digester (Jeníček et al., 2017). Current studies on microaeration are primarily confined to demonstrating the positive effects of microaeration and lacking an in-depth discussion on underlying mechanisms, which is crucial for better process optimization, full-scale application and maturation of the microaerated AD technology.

Recently, several comprehensive reviews on effects of oxygen on AD processes were published (Botheju and Bakke, 2011; Girotto et al., 2016; Krayzelova et al., 2015). However, in-depth discussion on the mechanisms, microbial metabolism, bioenergetics and pathway of microaerated AD process is still lacking.

This review critically examines the fundamental aspects of anaerobic digestion under microaerobic condition and synergistic interaction among microbial communities that co-exist and thrive in such environment. Furthermore, important considerations for designing and operating a microaeration-based AD system are discussed. The effects of microaeration on microbial community structures and functions along with enzymatic, bioenergetics, physiological and biochemical information are critically reviewed. Future perspectives and research gaps on microaeration-based AD system are also highlighted.

2. Microaerobic environment

2.1. Definition of microaerobic environment

Microaeration is the dosing of small amount of air or oxygen (typically ranged from 0.005 to 5 L O_2/L reactor/day) into an anaerobic system. Different homologous terminologies are being used, namely micro-aeration, limited aeration, micro-oxygenation, oxygenation, and microaerobic condition among others (Krayzelova et al., 2015). Air or oxygen could be dosed either one time, intermittently (pulse-mode) or continuously at different stages of the AD process (pretreatment, during digestion or post digestion) (Girotto et al., 2016). As a result, the terminology “microaeration” has been vaguely defined due to lack of standard method to precisely monitor oxygen level in the AD system.

The most practical and sound approach to standardize the microaerobic condition among the reported studies, is to use redox potential or oxidation-reduction potential (ORP). ORP is a measure of tendency of an aqueous solution to accept or donate electrons (APHA-AWWA-WEF, 2005) and is commonly used to measure the oxidative (more positive values) or reductive (more negative values) environment of in aqueous phase. Since ORP varies linearly with the logarithmic of oxygen concentration, the ORP electrode can precisely sense very small changes in oxygen concentration in the aqueous phase. For example, ORP value of −50 mV corresponds to the detection limit of DO probe at 0.1 mg/L (3 μM) and ORP values become more negative as DO reduces further (Ndegwa et al., 2007). A microaerobic condition may be defined as an aqueous media with ORP value in the range of 0 to −300 mV with reference to standard hydrogen reference electrode (SHE). Anaerobic environment has an average ORP level of −300 mV or lower (Khanal and Huang, 2006; Krayzelova et al., 2015). The ORP values of > 0 mV represent oxidizing environment of the aerobic condition (Fig. 1).

Facultative microorganisms grow under both anoxic and oxic conditions with some strains performing aerobic respiration even at nanomolar O_2 concentration (DO > 3 nM) (Morris and Schmidt, 2013; Stolper et al., 2010). Although aerotolerant anaerobes and obligate anaerobes are not considered facultatives, they co-exist and co-metabolize with facultatives under microaerobic condition with various

antioxidative mechanisms, which are discussed in the Section 2.3. As depicted in Fig. 1, facultatives grow optimally and produce fermentation products like ethanol, butanol and VFAs within the ORP range of 0 to −300 mV. In addition, the common redox couples of the reactions in AD processes such as SO_4^{2-}/HS^- (sulfidogenesis) or CO_2/CH_4 (methanogenesis) also have the standard redox potential ($E^{o\prime}$) in this microaerobic range of −200 to −300 mV. The dosing of air or oxygen to an anaerobic process elevates the ORP to a more positive value, which then slowly decreases with the consumption of injected O_2 by facultative bacteria. Several studies employed ORP as the controlling parameter for microaeration to enhance VFAs production (ORP of −100 to −200 mV) (Yin et al., 2016), sulfide removal (ORP of −275 to −265 mV) (Khanal et al., 2003), and to optimize fermentation process for ethanol, butanol, and propanediol production (ORP of −50 to −350 mV) (Liu et al., 2013). By manipulating the extracellular ORP of the environment, intracellular metabolic pathway of microbes (anaerobic oxidation, fermentation, or aerobic oxidation) can be modified at the molecular levels via electron flow, reducing power (i.e., NADH concentration), and gene expression (Celińska, 2010; Liu et al., 2013). As a result, ORP could be used as an effective parameter for monitoring and controlling the microaeration in an AD system. To standardize comparison among different studies, ORP values should be reported in mV with SHE as pointed out in this review.

2.2. Metabolism and energy in microaerobic environment

Microaeration-based AD processes create a unique environment that overlaps between anaerobic and aerobic conditions by maintaining niches for both anaerobes and micro-aerobes in such environment.

Under anaerobic condition, organic matters are partially reduced to energy-rich compounds (consist of phosphate bond or coenzyme A (CoA) molecule), which generate ATP via substrate level phosphorylation. Due to the lack of an effective mechanism to transport electrons, anaerobes produce less energy from the organic degradation reactions (4 mol ATP/mol glucose) (Fig. 2). In addition, to maintain the redox balance (ratio of $NADH/NAD^+$) under an anaerobic condition, fermentative products (e.g., ethanol, lactate, and VFAs) are produced to regenerate NAD^+ from NADH (Madigan et al., 2015). These intermediates are further converted to CH_4 via syntrophic oxidation reactions coupled with methanogenesis while generating minimal amount of energy using ion pumps to create proton or sodium motive force for ATP generation (Fig. 2).

Under aerobic condition, on the other hand, acetyl-CoA goes through tricarboxylic acid (TCA) cycle to be completely oxidized to CO_2 through highly energetic reaction. Using O_2 as a terminal electron acceptor, aerobic oxidation of 1 mol glucose generates total of 32 mol ATP via glycolysis, TCA cycle and oxidative phosphorylation of reduced coenzymes (NADH and $FADH_2$) (Fig. 2). With this highly energetic metabolism (Table 1), aerobic microbes rapidly consume organic matters and grow faster than anaerobes, as evident from higher maximum specific growth rate, substrate affinity, and biomass yield (Table 2).

Table 1 shows the comparison between energy produced from different VFAs using aerobic respiration and syntrophic anaerobic oxidation. Thermodynamic constraint of syntrophic anaerobic oxidation of VFAs to acetate and H_2 is a bottleneck in AD processes leading to VFAs accumulation and process instability, especially at high OLRs (Leng et al., 2017). Therefore, microaerobic condition integrating the aerobic VFAs oxidation by heterotrophs with anaerobic methanogenesis could be a promising strategy to facilitate energetic conversions of intermediates to maintain overall stability of AD processes. To couple aerobic oxidation with anaerobic reduction reactions, effective microaeration control strategy is needed to prevent inhibition of obligate anaerobes, as discussed in Sections 2.3 and 3.

2.3. Mechanisms of oxygen tolerance in microaerobic conditions

In microaerobic environment, facultative microbes consume and partially reduced oxygen molecules, generating reactive oxygen species (ROS) (i.e., O_2^-, H_2O_2, and •OH) as shown in Fig. 3. These active radicals and ions are highly oxidative and can damage cell membrane, protein and DNA of microorganisms (Ezraty et al., 2017). Aerobic and facultative bacteria can produce anti-oxidative enzymes and neutralize these oxidative species, allowing them to thrive in an aerobic condition (Fu et al., 2015; Imlay, 2013). As shown in Fig. 3, the common anti-oxidative enzymes are superoxide dismutase (SOD), expressed by gene *sod*; superoxide reductase or rubredoxin oxidoreductase; catalase, expressed by gene *kat* or cytochrome-*c* peroxidase expressed by gene *ccpA*; and alkyl hydroperoxide reductase, expressed by gene *ahp*. However, recent studies reported that under microaerobic condition, several strict anaerobes could also produce these enzymes to adapt to the oxidative environment. For example, under an oxidative stress, anaerobes *Clostridium perfringens, Clostridium acetobutylicum, Bacteroides fragilis, Desulfovibrio gigas, Methanosarcina barkeri, Methanobacterium* and *Methanobrevibacter* showed 2 to 30 times overexpression of genes encoding SOD, catalase, and superoxide reductase enzymes as compared to anaerobic condition without added oxidants such as H_2O_2 or O_2 (Brioukhanov et al., 2006; Brioukhanov and Netrusov, 2007; Horne and Lessner, 2013). These microorganisms could survive after exposure of up to 500 μM H_2O_2, 1 h of aeration, and O_2 in headspace at 5% *v*/v, or DO up to 5.6 mg/L (Brioukhanov and Netrusov, 2007). New species of acetoclastic methanogens were also discovered in microaerobic soils, plausibly contributing to methane generation in wetland (Angle et al., 2017). Such adaptive responses of strict anaerobic bacteria and archaea allow higher tolerance of these microorganisms to the microaerobic environment.

Co-existence and synergistic interaction between facultative and anaerobic microorganisms in AD process is another aerotolerant mechanism of strict anaerobes that cannot produce anti-oxidative enzymes. In AD process, microbial consortia tend to form flocs with strict anaerobes at the inner core and facultative bacteria at the outer layer (Botheju and Bakke, 2011; Liu et al., 2009). Facultative bacteria (i.e., hydrolytic and fermentative bacteria) with higher anti-oxidative enzyme activity could scavenge ROS and protect the anaerobes, which are more susceptible to oxygen (Fig. 4). In addition, oxygen and ROS concentrations decrease across the bioflocs layer thereby protecting strict anaerobic microorganisms located at the inner core of the flocs from oxidative stress (Stewart and Franklin, 2008). As depicted in Fig. 4, the synergetic relationship is also shown in substrate flow from outer to inner section of the aggregated flocs, where hydrolytic bacteria consume oxygen to breakdown complex organic matters into various intermediates that serve as substrates for fermentative bacteria and then for methanogens. As a result, the residual oxygen, which can be measured by ORP, declines to the level that is within the tolerance limit of strict anaerobes (Fig. 4c). Hence, the outer layer of facultative bacteria possibly acts as both physical and biological shield against oxygen thereby protecting strict anaerobic bacteria and archaea, and allowing them to survive and function effectively under microaerobic condition.

On the other hand, over aeration beyond the oxygen consumption and anti-oxidation capacities of facultative bacteria, can cause detrimental effects on strict anaerobes due to high concentration of ROS that damage their phospholipid membrane and DNA (Xu et al., 2014; Fig. 4b). In two-stage reactor configuration, when the hydrolytic reactor is microaerated, the risk of inhibiting methanogens would be minimal (Xu et al., 2014). However, in single stage reactor configuration, facultative bacteria have been reported to quickly consume oxygen and protect the oxygen-sensitive methanogens and other obligate anaerobes (Botheju and Bakke, 2011).

Insufficient oxygen dosing, in some cases, may also cause adverse effect on AD system as it may alter the delicate balance of the microbial communities thereby requiring the microbes to adjust to the new

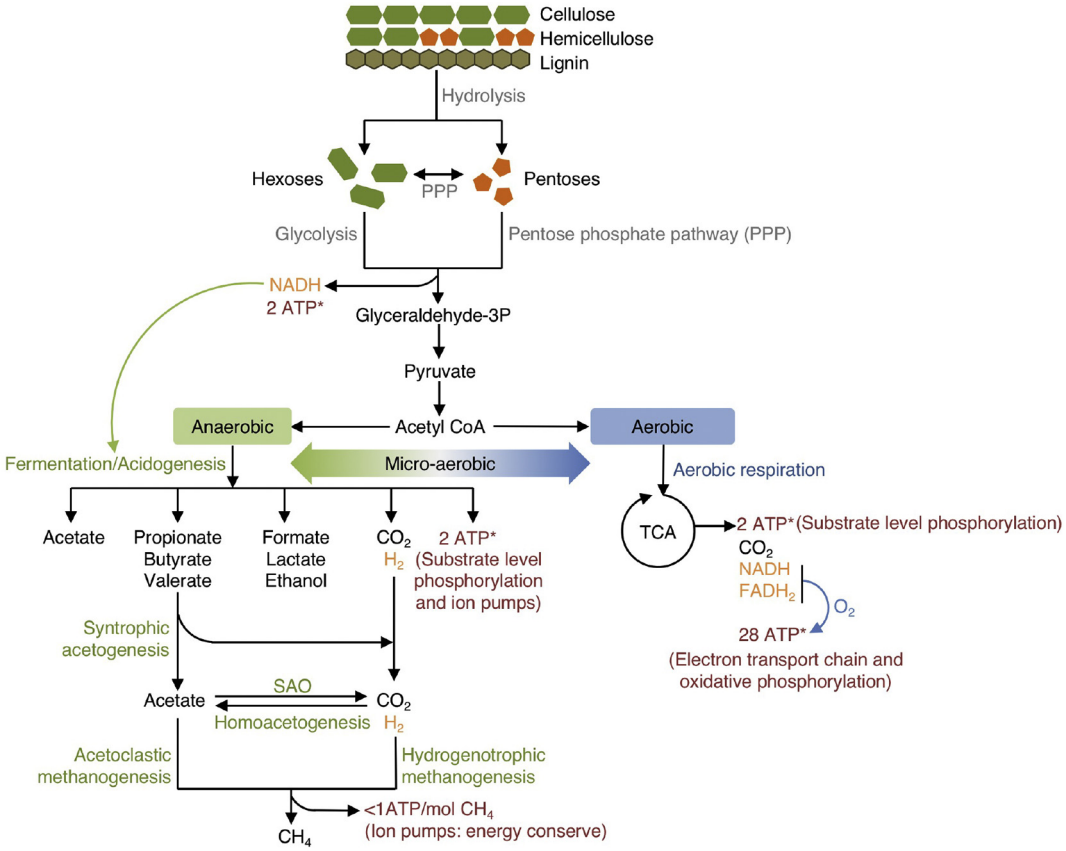


Fig. 2. Metabolic pathway and energy production from lignocellulosic substrate in anaerobic and microaerobic environments. *ATPs produce from 1 mol glucose. PPP: pentose phosphate pathway. SAO: syntrophic acetate oxidation. TCA: tricarboxylic acid cycle (i.e., citric acid cycle or Krebs cycle). Adapted from Madigan et al. (2015).

condition (Zhu et al., 2009). To supply sufficient amount of oxygen, various factors affecting oxygen transfer and utilization rate need to be considered, which are elucidated in the following Section 3.1.

3. Considerations for operating micro-aeration system

3.1. Factors affecting microaeration-based AD system

Microaeration rate (or intensity) is the main factor determining the effects of microaeration on AD process. The microaeration rate varies depending on specific purpose such as controlling VFAs accumulation, stabilizing the process, scavenging hydrogen sulfide, improving VFAs production, enhancing hydrolysis and overall methane yield (Fig. 5 and Table 3). For example, micro-aeration rate can be calculated based on O_2/S^{2-} molar ratio of 0.5–1.5 for removing H_2S (Krayzelova et al., 2015) or 2–10% of substrate COD for enhancing hydrolysis (Botheju et al., 2010).

Despite the fact that microaeration rates could be determined based on specific purpose, both oxygen transfer rate (OTR) and oxygen utilization rate (OUR) need to be considered to provide sufficient amount of oxygen (Fig. 5).

The OTR in the AD process depends on several factors such as

Table 1
Energetics of VFAs conversion reactions under microaerobic and anaerobic conditions.

Reactions	$\Delta G^{o\prime}$ (kJ/reaction)	Electrons transferred	$\Delta G^{o\prime}$ (kJ/e^-)
Anaerobic condition			
Valerate + 3 H_2O = 2.5 Acetate + 3 H_2 + 1.5H^+	72.6	6	12.1
Butyrate + 2 H_2O = 2 Acetate + 2 H_2 + H^+	48.3	4	12.1
Propionate +1 H_2O = 1.5 Acetate + H_2 + 0.5H^+	24.2	2	12.1
Acetate + H^+ + 2 H_2O = 2 CO_2 + 4 H_2	94.9	8	11.9
Microaerobic condition			
Valerate + H^+ + 6.5 O_2 = 5 CO_2 + 5 H_2O	−2773.6	26	−106.7
Butyrate + H^+ + 5 O_2 = 4 CO_2 + 4 H_2O	−2133.8	20	−106.7
Propionate + H^+ + 3.5 O_2 = 3 CO_2 + 3 H_2O	−1493.8	14	−106.7
Acetate + H^+ + 2 O_2 = 2 CO_2 + 2 H_2O	−853.9	8	−106.7
Methanogenesis			
Acetate + H^+ = CO_2 + CH_4	−35.9	8	−4.5
CO_2 + 4 H_2 = 2 H_2O + CH_4	−130.8	8	−16.4

Note: *Standard Gibb's free energy ($\Delta G^{o\prime}$) for all gases is calculated at 1 atm, VFAs and H_2O at 1 M, H^+ concentration at pH 7.0, and at 25 °C (Dolfing, 2015; Thauer et al., 1977). Values of $\Delta G^{o\prime}$ when HCO_3^- + H^+ is used instead of CO_2 + H_2O for balancing the reactions (Thauer et al., 1977).

Table 2
Biokinetic parameters for bacterial and archaeal groups.

	μ_m (1/day)	K_S (mg COD/L)	q_m (g COD/g VSS/day)	Y (g VSS/g COD)	Reference
Aerobic heterotrophic bacteria	13.20	20	27	0.49	(Rittmann and McCarty, 2001)
Carbohydrate fermentative bacteria	1.20	500	9.8	0.13	(Ni et al., 2015)
Anaerobic acetate degrader	0.38	165	9.5*	0.04	(Lawrence and McCarty, 1969)
Anaerobic propionate degrader	0.31	60	7.8*	0.04	(Lawrence and McCarty, 1969)
Anaerobic butyrate degrader	0.35	13	7.0*	0.05	(Lawrence and McCarty, 1969)
Sulfide-oxidizing bacteria	1.4	–	5[a]	0.28[c]	(Rittmann and McCarty, 2001)
Sulfate-reducing bacteria	0.29	–	1.05[b]	0.28[d]	(Rittmann and McCarty, 2001)
Acetoclastic methanogens	0.30	–	8.4	0.035	(Rittmann and McCarty, 2001)
Hydrogenotrophic methanogens	0.50	–	1.1[b]	0.45[d]	(Rittmann and McCarty, 2001)

Note: μ_m: maximum specific growth rate (1/day), K_S: half-velocity constant (mg chemical oxygen demand (COD)/L), q_m: maximum specific substrate utilization rate (g COD/g volatile suspended solids (VSS)/day), Y: biomass yield (g VSS/g COD). Specific units: [a] g S/g VSS/d, [b] g H_2/g VSS/d, [c] g VSS/g S, [d] g VSS/g H_2.
* Calculated from equation $\mu_m = q_m$*Y (Rittmann and McCarty, 2001).

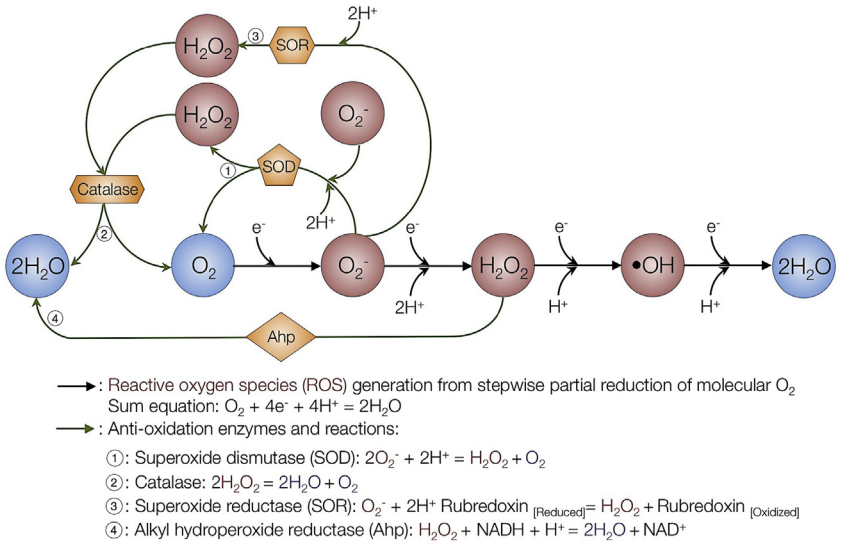


Fig. 3. Reactive oxygen species generation and anti-oxidative stress mechanism of microorganism. (Adapted from Green and Paget (2004)).

reactor configuration, microaeration method (i.e., use of air or oxygen, bubble size, injection in aqueous or gaseous phase), TS content of substrate in the reactor, etc. (Garcia-Ochoa and Gomez, 2009). For example, injection of air or oxygen in the reactor headspace has been widely practiced for the removal of hydrogen sulfide from biogas with removal efficiency as high as 99% (Krayzelova et al., 2015). However, due to air/oxygen diffusion limitation at gas-liquid interphase, especially at high solids conditions, this microaeration method failed to remove dissolved sulfide in liquid phase or improve hydrolysis rate (Boe et al., 2010; Sheets et al., 2015). Novel microaeration technique using electrolysis to supply oxygen and hydrogen to AD reactor was capable of scrubbing hydrogen sulfide from biogas, enhancing hydrolysis and improving methane yield via hydrogenotrophic pathway (Tartakovsky et al., 2011). Microaeration can be carried out by injecting air/oxygen into the inoculum tank, hydrolytic reactor of a two-staged reactor configuration, directly into the AD reactor, into the biogas or sludge recycling line, or into digestate storage tank in order to achieve different objectives (Girotto et al., 2016). Studies have shown that the activities of methanogens were even improved as compared to the strict anaerobic condition due to more diverse, well-balanced microbial communities as evident from low VFAs and stable pH (Fu et al., 2016; Lim et al., 2014).

The OUR in microaeration-based AD reactors depends on inoculum used and substrate fed to the reactor. Substrate types and loading rates affect the optimum microaeration rate (Table 4). For example, microaeration intensity (L O_2/L reactor/day or L O_2/g TS) and treatment duration to enhance hydrolysis of recalcitrant substrates such as lignocellulosic biomass is higher than that of food wastes, high strength wastewater or sewage sludge (Díaz et al., 2011; Jagadabhi et al., 2010; Sawatdeenarunat et al., 2017). Microaeration for the substrates that have higher hydrolysis rate requires a precise oxygen dosing control to prevent the excess accumulation of VFAs in the system and/or facultative consumption of hydrolyzed products while achieving enhanced methane yield (Johansen and Bakke, 2006; Lim and Wang, 2013; Xu et al., 2014). In all applications of microaeration, substrate competition between facultative bacteria and anaerobic methanogens needs to be considered to achieve enhanced digestion while preventing overall reduction in methane yield due to consumption of substrates by facultative bacteria.

Since different microbial consortia utilize free oxygen at different rates, interaction between microaeration rate and inoculum source (i.e., between OTR and OUR) needs to be studied for microaeration optimization (Tsapekos et al., 2017). By better understanding of OTR and OUR in microaerobic condition, the microaeration-based AD process could be scaled up while maintaining the efficiency comparable to that of lab-scale AD process (Fernández-Sandoval et al., 2017). However, OTR and OUR in the microaerobic condition has yet to be critically examined. The best mathematical tool available to model the effect of oxygen on anaerobic process is the ADM1-Ox – an adapted model from the well-known Anaerobic Digestion Model No. 1 (ADM1) (Batstone et al., 2002; Botheju et al., 2010).

Based on the purpose of microaeration, Table 3 could be used along with Fig. 5 as a guidance to determine the microaeration rate at the specific reactor conditions.

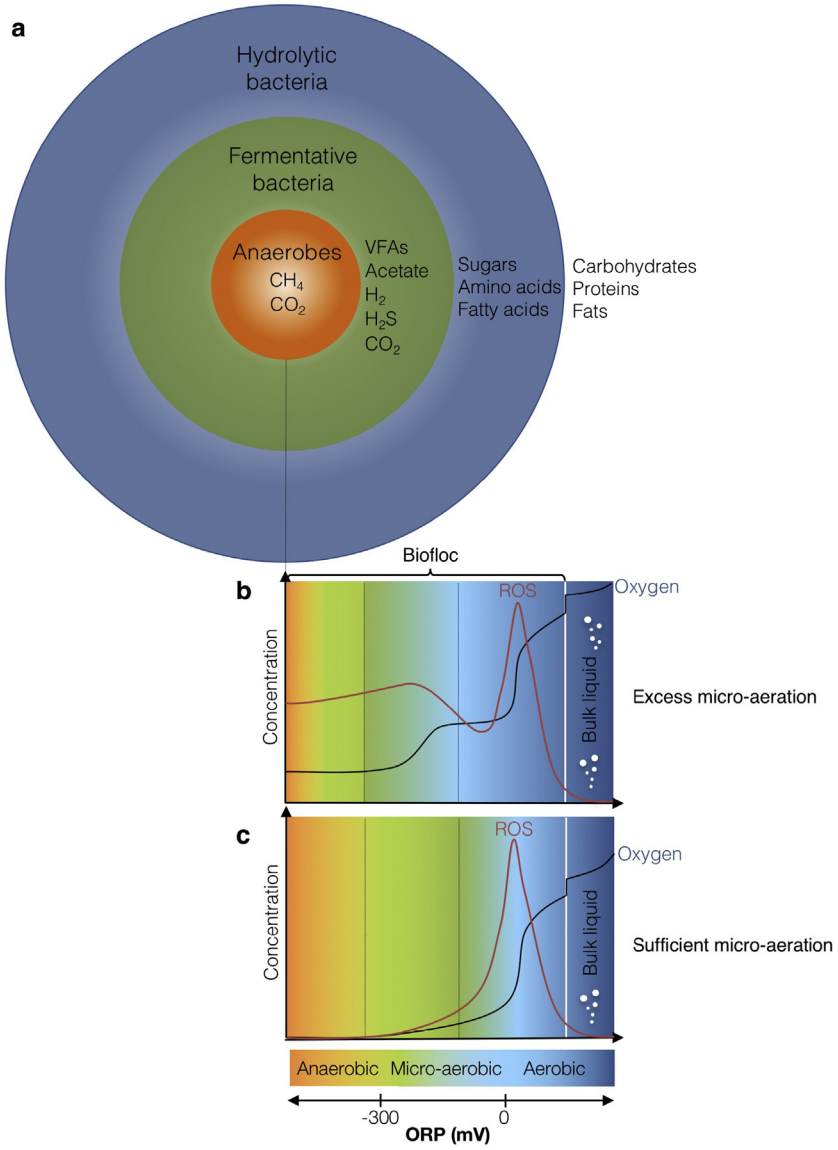


Fig. 4. Anti-oxidative stress mechanism of microbial community in microaerobic environment. Distribution of various microbial groups in bioflocs: strict anaerobes aggregate in the center core, hydrolytic bacteria at the outer layer, and fermentative bacteria spread across the biofloc. Substrate gradient across bioflocs where products from the outer group are substrate to the inner group (a), gradient of oxygen and ROS across the bioflocs in excess microaeration (b), and sufficient microaeration intensity (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Process control for microaeration

Process control is critically important to allow precise control of microaeration in AD system. Besides time-based on-off control, other fine process control systems such as proportional-integral-derivative (PID) control (Liu et al., 2011) and ORP-based control (Chang et al., 2014; Khanal and Huang, 2006; Nghiem et al., 2014) are recommended. The use of ORP as a control parameter could maintain enough residual oxygen without inhibiting obligate anaerobes, since dosing of oxygen could be controlled precisely based on oxygen consumption rate of facultative bacteria in the system. Application of ORP for process monitoring and control of fermentation process for producing desired products by sensing intracellular metabolic profile was also implemented in microaerobic and anaerobic environments (Liu et al., 2013). For these reasons, ORP-based process control system appears to be the most appropriate technique for controlling microaeration. However, ORP is also susceptible to environmental factors, such as pH, temperature, and ionic strength, which could be auto-compensated through automatic monitoring and control system such as supervisory control and data acquisition (SCADA) system.

Selection between air and pure oxygen for dosing into the anaerobic reactor should be based on the purpose of the microaeration and the associated cost. If the sole purpose of microaeration is to increase hydrolysis and VFAs production instead of biogas production, air should be used for microaeration. If methane production is the main purpose, the concern of biogas dilution with nitrogen from injected air (even though at minimal level) could be minimized by effective microaeration control. Díaz et al. (2015) suggested the application of oxygen at concentration of 95% (by volume), which was generated from pressure swing adsorption generator as a more economical substitute for air or pure oxygen.

The challenge associated with automatic process control of microaeration is the bottleneck of the implementation of microaeration-based full-scale AD system. The development of an automatic microaeration dosing system with intrinsic control and long-term stability is required to make this technology successful.

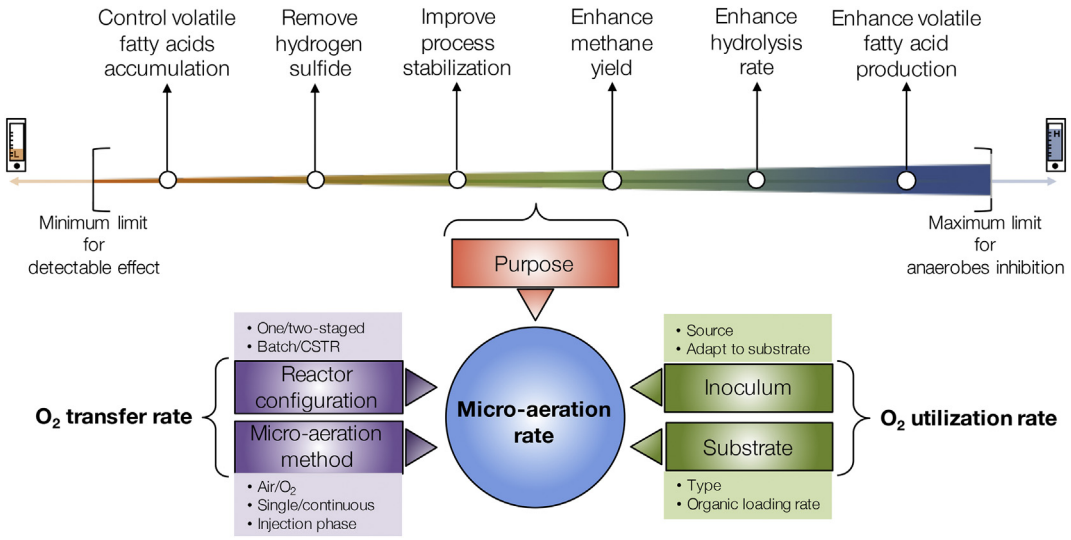


Fig. 5. Microaeration rate for various purposes and factors for consideration.

4. Syntrophic relationship among the microbial communities in microaerated AD system

In AD process, complex organic matters are broken down through a series of interdependent steps, in which products of the previous step serve as substrates for the following step (Fig. 6). As a result, microorganisms interact with each other in such a way that it allows perpetual substrate flows from hydrolysis to fermentation and then to methanogenesis. This syntrophic relationship between bacterial and archaeal communities allows them to function effectively even with limited ATP production from the thermodynamically less favorable reactions (Stams and Plugge, 2009).

Nevertheless, the balance and syntrophic relationships among different microbial groups have to be maintained in microaerobic AD process for stable operation. The microaeration-based AD system has to maintain the balance between anaerobic and aerobic metabolisms, between oxidation and reduction reactions, between OTR and OUR, and between substrates and products.

In order to achieve such balance, synergies between niches of hydrolytic and fermentative bacteria, methanogens and syntrophs, and sulfate-reducing bacteria and sulfide-oxidizing bacteria need to be maintained.

4.1. Hydrolytic and fermentative bacteria

Taxonomic analysis of 21 full-scale AD plants showed that *Bacteria* domain consisted of > 80% of total DNA sequences (Sundberg et al., 2013). Among bacteria, *Firmicutes*, *Proteobacteria*, *Bacteroidetes*, and *Actinobacteria* were the dominant bacterial phyla. However, this microbial structure in AD processes varied depending on inoculum types (Yin et al., 2016), reactor configurations (Lim and Wang, 2013), stages of operation (i.e., startup or stable phase) (Goux et al., 2016), and substrate types (Hanreich et al., 2013). Metaproteomic study found out that *Fimicutes* and *Bacteroidetes* phylum are responsible for the hydrolysis of hemicellulose, cellulose and other polysaccharides, while *Proteobacteria* are mainly glucose and VFAs utilizing bacteria (Ariesyady et al., 2007; Hanreich et al., 2013).

Microaeration facilitates the establishment of more diverse hydrolytic and fermentative bacterial communities with higher activity in AD systems (Fu et al., 2016; Lim et al., 2014). Enhancement of diversity and activity of these facultative bacteria is the foundation for controlling the VFA concentration (either enhance or reduce), which promotes the overall stability of the AD process. Compared to strict AD system, microaerated AD system consisted of a higher proportion of *Firmicutes* phylum that is associated with higher substrate hydrolysis rate (Table 5). Hydrolysis is reported to be the rate-limiting step in AD of high solids organic substrates such as municipal solid wastes and lignocellulosic biomass (Ren et al., 2009; Shrestha et al., 2017). Microaeration is a promising environment friendly biological pretreatment technique to enhance methane yield from such recalcitrant substrates. The increase in extracellular hydrolytic enzymes (e.g., amylase, protease and cellulase) production from a more abundant and diverse hydrolytic bacterial communities under microaerobic condition enhances the hydrolysis of carbohydrates, proteins, and other complex organic substrates (Lim et al., 2014; Zhu et al., 2009). For example, *Firmicutes* population shifted from 58% to 72% after microaeration treatment (Lim et al., 2014). Upon microaeration, augmentation of bacterial activity in phylum *Firmicutes* (especially in *Clostridia* and *Bacilli* class) resulted in a 3-folds increase in acetic and butyric acids concentration, which ultimately resulted in higher methane yield (Xu et al., 2014). Johansen and Bakke (2006) also reported up to 60% enhancement in hydrolysis after 4 days of microaerobic pretreatment. Similar trends were also found in recent studies with various substrates, such as municipal solid waste (Nguyen et al., 2007), vegetable waste (Zhu et al., 2009), food waste (Xu et al., 2014), grass silage (Jagadabhi et al., 2010), sugarcane bagasse (Fu et al., 2015), co-digestion of food waste and domestic wastewater (Lim and Wang, 2013), and Napier grass (Sawatdeenarunat et al., 2017).

Augmentation in hydrolytic activity results in higher soluble substrate availability for fermentative bacteria to produce VFAs and other fermentative products, which could serve as potential substrates for producing high value alcohol-based fuels, biopolymer, and electricity using microbial fuel cell (Botheju and Bakke, 2011; Jagadabhi et al., 2010; Sawatdeenarunat et al., 2016). As a result, microaerobic AD process could be used to produce VFAs as the final products instead of biogas; and since the strict anaerobic methanogens are excluded from the process, there is much greater flexibility to control microaeration in the acid-producing digester. Under microaerobic condition, facultative anaerobes were more likely to switch their intracellular metabolic pathway from anaerobic fermentation to aerobic respiration, which is much more thermodynamically (Table 1) and kinetically favorable (Table 2). Central regulators such as fumarate nitrate reduction (FNR) and anoxic redox control (AcrA/ArcB) upregulate the expression of genes involved in anaerobic metabolism (Forster and Gescher, 2014). Free O_2 under microaerobic condition can inactivate these central regulators, leading to the expression of genes related to aerobic metabolism. For example, deletion of *arcA* and *fnr* genes encoding these two regulators caused overexpression of genes related to TCA cycle such as

Table 3
Microaeration rate of various applications.

Objective	Reactor configuration and working volume	Substrate	Microaeration intensity	Dosing method	Oxygen dosing rate equivalent* (L O_2/$L_{reactor}$/d)	Result	Reference
Enhance hydrolysis	CSTR (3 L)	Food waste and brown water	5 mL and 7 mL O_2/L/d	Single dosing	0.005 and 0.007	More diverse bacterial population and higher VFA concentration	(Lim et al., 2014)
Enhance hydrolysis	CSTR (0.25 L)	Food waste and brown water	0.037 L O_2/L/d	Daily dosing (4 days)	0.04	Enhance hydrolysis and acidogenesis	(Lim and Wang, 2013)
Enhance hydrolysis	CSTR (0.5 L)	Primary sludge	500 mL air/d at flowrate 15 s/min	Semi-continuous time based	0.21	50–60% enhance hydrolysis. Reduced methane yield, VFAs and sCOD due to aerobic substrate consumption	(Johansen and Bakke, 2006)
Enhance hydrolysis	CSTR (1.6 L)	Primary sludge	2.5 mL air/min	Continuous	0.5	Increase solubilization of COD by hydrolysis of carbohydrates and protein	(Diak et al., 2013)
Enhance hydrolysis	CSTR (2.5 L)	Waste activated sludge	0.4 volume of air per volume of reactor per minute (vvm) and 0.08 vvm of air	Continuous	24–121	Increase solubilization	(Hasegawa et al., 2000)
Enhance hydrolysis	Leach bed reactor (4.6 L)	Synthetic food waste	129, 258, 387 L air/kg TS/d at flow rate of 1 L/min	Semi-continuous time based	2.1, 4.4, and 6.5	Middle aeration rate is best: increased hydrolysis and no effect on methanogenic reactor	(Xu et al., 2014)
Enhance hydrolysis and acidification	Leach bed reactor (260 L)	Municipal solid waste	1 L air/min for 2 h and stop 4 h. 3.2 Lair/kg TS/d	Semi-continuous time based	0.39	No significant increase of hydrolysis	(Nguyen et al., 2007)
Enhance methane yield	Batch (0.2 L)	Corn straw	12.5, 25, 50, 100 mL air/L/d	Single	0.003–0.021	Enhance methane yield and VS removal at lower micro-aeration intensity. Enhance diversity of phylum *Firmicutes*	(Fu et al., 2016)
Remove H_2S	CSTR (200*L*)	Waste activated sludge	0.013–0.024 L O_2/L/d	Continuous	0.013–0.024	99% H_2S removed. No negative effect on methane yield	(Fdz-Polanco et al., 2009)
Remove H_2S	Fluidized bed (1.7 L)	Vinasse	0.7–0.9 L air/L/d (Molar ratio of O_2/S is 8–10)	Continuous	0.15–0.19	Complete removal of H_2S in biogas. No aerobic respiration of organic substrate	(van der Zee et al., 2007)
Remove H_2S	UASB (10.5 L)	Pulp mill wastewater	3–6 mL air/L/min (0.38 mL O_2/mg S)	Continuous	0.9–1.8	Up to 30% of H_2S removal. 40–80% enhance COD removal rate.	(Zhou et al., 2007)
Remove H_2S	CSTR (50 L)	Primary sludge	0.14 mL O_2/s to maintain ORP between −320 to 270 mV	Semi-continuous ORP based	na	99% H_2S removal. No effects on VS and COD removal rate	(Nghiem et al., 2014)
Remove H_2S	Sludge digester (200 L)	Waste activated sludge	0.25 L O_2 / L sludge	Continuous	0.01	98% H_2S removal from biogas.	(Díaz et al., 2011)
Remove H_2S	UASB (2.7 L)	Synthetic brewery wastewater	0.5 mol O_2/S^{2-} (1 L air/d)	Continuous	0.08	73% H_2S removal.	(Krayzelova et al., 2014)
Remove H_2S	Sludge digester (200 L)	Waste activated sludge	0.25 L O_2/L feed and 1.27 L air/L feed	Continuous	0.25	99% H_2S removal.	(Dı´az et al., 2010)
Control VFAs accumulation and improve effluent quality	CSTR (10 L)	Waste activated sludge	1.6 L /d	Continuous	0.03	Micro-aerobic reactor had 3.5 times lower VFAs, 33% lower of sCOD, lower foaming and better dewaterability as compared with anaerobic reactor	(Jenicek et al., 2014)
Overcome overloading and improve reactor stability	CSTR (200 L)	Waste activated sludge	4.4 NL O_2/m^3.d	Continuous	0.01	Overcome hydraulic overloading. Promoted growth of hydrogenotrophic methanogens.	(Ramos and Fdz-Polanco, 2013)
Control pH	CSTR (3 L)	Synthetic wastewater	0.1 and 1 g O_2/L.d at flowrate 0.9 L O_2/min	Semi-continuous time- based	0.07 and 0.7	Help recover pH to neutral. Good for startup reactor.	(Zitomer and Shrout, 1998)
Produce VFAs	Leach-bed reactor (1 L)	Grass silage	1 L and 4 L air/min	Semi-continuous time based	0.5 and 5	Increased the VFAs production by 4 times. Extensive aeration reduced VFAs production	(Jagadabhi et al., 2010)
Produce VFAs	Batch (0.16 L)	Napier grass	15 and 30 mL O_2/gVS	Single	0.09 and 1.9	Highest VFAs production with 15 mL O_2/g VS and 3 days incubation time with cattle manure inoculum	(Sawatdeenarunat et al., 2017)

Note: CSTR: continuous-stirred tank reactor; UASB: up-flow anaerobic sludge blanket; sCOD: soluble chemical oxygen demand; VS: volatile solids; TS: total solids.
* Calculated from reported micro aeration intensity with assumption O_2 = 21% (v/v) of air.

Table 4
Factors determining the microaeration rate.

Factors	Microaeration purpose	Recommended microaeration rate[a]
Reactor configuration		
CSTR	Control VFAs accumulation Remove sulfide	Low
UASB	Enhance methane production Remove sulfide	Medium
Leached bed reactor	Enhance hydrolysis	High
Acid tank of two-staged reactor	Enhance VFAs production Enhance hydrolysis	High
Microaeration method		
Gas phase injection	Remove H_2S from gas phase	High
Liquid phase injection	Enhance hydrolysis Remove dissolved sulfide Enhance VFAs production Control VFAs accumulation	Low-Medium
Single injection	Enhance hydrolysis Enhance VFAs production	High
Intermittent/continuous injection	Control VFAs accumulation	Low
Inoculum		
Low biomass concentration	Reduce start-up time	Low
High biomass concentration	Control VFAs accumulation Enhance hydrolysis	High
Low biomass diversity	Enhance hydrolysis Control VFAs accumulation	High
High biomass diversity	–	Low
Substrate		
Recalcitrant	Enhance hydrolysis	High
Easily biodegradable	Enhance VFAs production Enhance methane yield	Medium
Low organic loading rate	Enhance hydrolysis	Low
High organic loading rate	Control VFAs accumulation	High

[a] Recommended microaeration rate based on previous studies: Low: 0.005–0.01, medium: 0.01–0.2, high: 0.2–5.0 L O_2/ L reactor/day.

sucC (succinyle-CoA synthase) and *sdhC* (succinate dehydrogenase) as compared to wild type (Shalel-Levanon et al., 2005).

Both the yield and composition of VFAs strongly depend on the substrate types and operating conditions including temperature, pH, OLRs, hydraulic retention time, and solids retention time (Sawatdeenarunat et al., 2017; Surendra et al., 2015). Thus, to optimize AD process to maximize VFAs yield, the above stated operating conditions should be maintained at the optimum range. During acidogenesis, the optimum range of ORP to facilitate VFAs production is reported to be between −100 and − 200 mV (Yin et al., 2016). At ORP higher than −100 mV, propionic acid was reported to be the dominant VFA (Wang et al., 2006). The shift in fermentation pathway such as increase in butyric acid production from conventional acetic acid producing pathway was achieved under microaerobic condition (Lim et al., 2014). These results can be applied to control ORP level using microaeration to enhance the yield of specific VFAs. However, molecular and multi-omic analyses need to be conducted to better understand the shift in metabolic pathway under microaerobic condition.

Microaeration promotes the diversity, growth and activities of rapid-growing facultative heterotrophs which would prevent the accumulation of VFAs in AD processes operating at high OLRs (Guo et al., 2015). For example, under microaerobic condition, propionic, butyric and valeric, and lactic acids were converted into acetic acid more efficiently (Lim and Wang, 2013). In addition, hydrogenotrophic methanogenic pathway was promoted under microaerobic condition that maintained low hydrogen partial pressure thereby maintaining stable pH of the reactor while efficiently producing methane (Ramos and Fdz-Polanco, 2013).

Lower effluent COD, better sludge dewaterability and significantly lower foaming problem of an AD process, which resulted from enhancement in hydrolysis rate and reduction of filamentous bacteria, were also reported during long-term operation under microaerobic condition (Jenicek et al., 2014). Nghiem et al. (2014), however, observed no significant differences in pH, VS removal, alkalinity, and methane production between anaerobic and microaerobic conditions. This could be due to rapid oxygen consumption by SOB for sulfide removal in which hydrogen sulfide in biogas decreased from over 6000 ppm to 30 ppm. Thus, the fate of oxygen in microaerobic system strongly depends on synergetic activities and substrate (oxygen) competition among different microbial groups.

In conclusion, hydrolytic and fermentative bacteria, as the most abundant microbial group in microaerobic condition, improve the AD process in multiple ways. The enhancement in diversity and activity of these bacteria with the availability of limited oxygen is the key for the operation of an efficient AD system. With the controlled dosing of appropriate amount of oxygen, improvement in VFAs production and methane yield, and overall system stability could be achieved.

4.2. Methanogens and syntrophs

Even though *Archaea* domain usually consists of just < 4% of microbial population, metagenomic study has shown that around 30% of protein released in AD process belongs to this domain (Hanreich et al., 2013). Acetoclastic *Methanosaeta, Methanosarcina* and hydrogenotrophic *Methanosarcina, Methanoculleus, Methanobacteriales,* and *Methanospirillum* are the main methane-producing archaea with the typical higher proportion of acetoclastic methanogens (Lim and Wang, 2013; Tang et al., 2004). However, in AD processes with high concentrations of VFA, ammonia and high temperature (55 °C, thermophilic AD), hydrogenotrophic methanogenesis is the dominant pathways (Hattorii, 2008). This could be due to the fact that hydrogenotrophic methanogens (HM) and syntrophic acetate-oxidizing bacteria (SAOB) are more resilient to inhibitors than the acetoclastic *Methanosaeta* (Demirel and Scherer, 2008; Wang et al., 2015).

Enhancement in hydrolysis and acidogenesis under microaerobic condition generates more substrate for methanogens that resulted in higher specific methanogenic activity and ultimate methane yield under microaerobic AD process (Fu et al., 2016; Lim and Wang, 2013). In addition, as discussed previously, the removal of sulfide and stabilization of pH in microaerobic process also create an optimum condition for the growth of methanogens. Microaeration also directly affects the methane production step by modifying the dominant methanogenic pathway. With the ability to use both acetate and hydrogen to produce methane, along with aerotolerance, *Methanosarcina* was found to be the dominant archaea in microaerobic system (Fu et al., 2016). The shift of dominant archaeal genus from *Methanosarcina* to *Methanoculleus* and from acetotrophic to hydrogenotrophic pathway following microaeration was reported in several studies (Ramos and Fdz-Polanco, 2013; Tang et al., 2004). The shift between acetoclastic and hydrogenotrophic pathways is highly dependent on the activity of SAOB (converting acetate to H_2 and CO_2) and homoacetogenic bacteria (reducing CO_2 to acetate) (Saini et al., 2011). *Thermacetogenium phaeum, Syntrophaceticus schinkii, Tepidanaerobacter acetatoxydans* and *Clostridium ultunense* are the common SAOB (Ariesyady et al., 2007). Dominance of SAOB and hydrogenotrophic methanogens was found in co-digestion of diverse substrates in full-scale AD systems in contrast to the acetoclastic

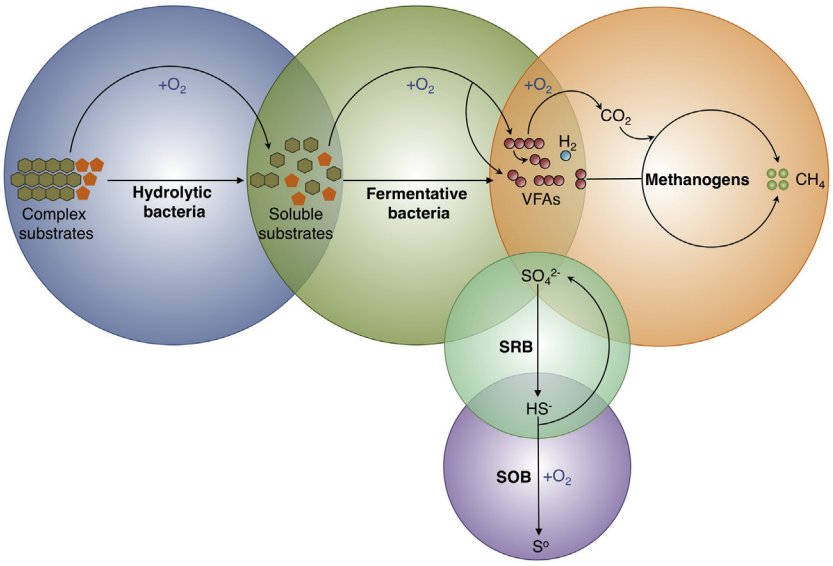


Fig. 6. Syntrophic relationship between diverse microbial communities in AD process under microaeration. SRB: sulfate-reducing bacteria. SOB: sulfide-oxidizing bacteria.

Table 5
Effect of microaeration on functional microbial groups.

Microaeration purpose	Effect on microbial group[a]	Microbial function	Reference
Enhance hydrolysis	*Firmicutes* (0.06-fold increase)	Hydrolytic bacteria	(Fu et al., 2016)
	Clostridiales (0.16-fold increase)	Hydrolytic bacteria grew on cellulosic substrate	
	Methanosarcina (1.1-fold increase)	Aerotolerant methanogens	
	Methanobacterium (1-fold increase)	Hydrogenotrophic methanogens	
Enhance hydrolysis	*Firmicutes* (0.24-fold increase)	Hydrolytic bacteria grew on various substrates	(Lim et al., 2014)
Produce VFAs	*Firmicutes* (1.28-fold increase)	Hydrolytic bacteria with extracellular enzymes	(Yin et al., 2016)
Remove sulfide	*Desulfovibrio psychrotolerans* (0.55 to 1-fold reduce)	Sulfate-reducing bacteria	(Chang et al., 2014)
Remove sulfide	*Arcobacter mytili*, *Thiobacillus, Halothiobacillus, Sulfuricurvum, Acidithiobacillus* (identified by DGGE gel)	Sulfide-oxidizing bacteria	(Díaz et al., 2011b; Ramos and Fdz-Polanco, 2013)

[a] Fold changes (increase or decrease) in relative abundance (%) as compared to the same process under anaerobic condition.

methanogens for AD process fed with sewage sludge (Sundberg et al., 2013).

In controlled microaerobic digestion system, effective pathway of converting VFAs to methane could be achieved with enhanced methane production. Under such microaerobic condition, facultative heterotroph could join the syntrophic relationship between methanogens and syntrophs as previously discussed. The traditional syntrophic interaction between SAOB and hydrogenotrophic methanogens could be replaced by the synergetic interaction between facultative bacteria (producing H_2 and CO_2) and hydrogenotrophic methanogens (consuming H_2 and CO_2) (Zabranska and Pokorna, 2018). The later combination is energetically more favorable (Table 1) and could promote the stability of AD processes operating at high OLRs, if microaeration is meticulously controlled. Therefore, changes in VFAs, H_2 and CO_2 concentration of AD processes operating under microaerobic condition can affect the archaeal communities and thus the dominant methanogenic pathway. This interesting concept needs to be examined to better understand the dynamic interaction between bacterial and archaeal communities using novel multi-omics approaches.

4.3. Sulfate-reducing bacteria and sulfide-oxidizing bacteria

Sulfate-reducing bacteria (SRB), such as *Desulfobacterium, Desulfovibrio, Desulfomicrobium,* reduce sulfate to a toxic and highly corrosive hydrogen sulfide (H_2S) in biogas and dissolved sulfide ($H_2S + HS^-$) in aqueous phase that inhibits anaerobes, especially methanogens and is corrosive to the pipes and metal parts of reactors. In addition, competition for substrates (acetate, H_2, and other intermediates) between SRB and methanogens is equally important to consider, especially when the goal is to enhance methane production (Fig. 6). SRB have higher substrate affinity for acetate and H_2 and have higher growth rate, and outcompete methanogens thereby resulting in low methane yield and high H_2S content in the biogas (Chen et al., 2014; Liamleam and Annachhatre, 2007; Muyzer and Stams, 2008).

Sulfide/sulfur-oxidizing bacteria (SOB) are responsible for biological oxidation of sulfide (HS^-) to elemental sulfur (S^o), sulfate (SO_4^{2-}) and thiosulfate ($S_2O_3^{2-}$) by producing Sox protein complex that catalyzes the sulfide oxidation reactions (Pokorna and Zabranska, 2015). Majority of SOB belong to phylum *Proteobacteria* with *Thiobacillus, Halothiobacillus*, *Sulfuricurvum* and *Acinetobacter* as the main genus (Krayzelova et al., 2015). In microaerobic condition, dominant SOB are chemolithotrophs, and use oxygen as an electron acceptor and CO_2 or/and organic compounds as carbon and energy source. Addition of oxygen was found to enhance SOB activity with over 99% removal of H_2S from biogas (Jenicek et al., 2011).

Co-existence of SOB and SRB occurs in bioreactor fed with sulfate-rich substrates, where SRB reduce SO_4^{2-} to sulfide and SOB consume O_2 to oxidize sulfide into elemental sulfur (S^o) and/or SO_4^{2-} (Fig. 6).

Under microaerobic condition, the oxygen-sensitive SRB normally exist at oxygen-depleted zone and are protected from oxygen toxicity by SOB activity (Stewart and Franklin, 2008). Some species of SRB with aerotolerance such as *Desulfovibrio oxyclinae*, were also reported to grow independently under oxic condition, while other species tend to co-exist with aerobic or facultative bacteria as an oxygen-defense mechanism (Muyzer and Stams, 2008). This interaction is pivotal in controlling the microaeration system, since at low sulfate concentration, the activities of SRB and SOB are also low due to limited substrate availability. Hence, the injected oxygen is not completely consumed by SOB thereby resulting in the inhibition of strict anaerobic SRB and methanogens (Khanal and Huang, 2003).

Air or oxygen is injected into reactor headspace and the holding time of biogas in headspace is the main factor governing the H_2S removal efficiency (Krayzelova et al., 2015). Several studies on microaeration of liquid phase of the anaerobic reactor were able to remove both H_2S in biogas and dissolved sulfide in liquid (Díaz et al., 2011b; Khanal and Huang, 2006). Microaeration rate for sulfide removal can be calculated using molar ratio of O_2/S^{2-}. Stoichiometrically, 0.5 mol O_2 per mol S^{2-} is required for converting H_2S into S^o. Microaeration rate, however, is affected by various factors, such as sulfide concentration, soluble substrate concentration, and pH among others. Therefore, a wide range of O_2/S^{2-} molar ratios (i.e., 0.15 to 2.0) have been reported in the literature (Krayzelova et al., 2015).

Although microaeration was able to remove over 99% of dissolved sulfide in liquid phase, pipe clogging associated with sulfur element deposits and methane yield reduction due to aerobic oxidation of substrate resulting from excess aeration, were reported as the major challenges of biological desulfurization using microaeration (Krayzelova et al., 2015). To overcome the pipe-clogging problem, the microaeration membrane dosing system has been examined which allows elemental sulfur to deposit/precipitate on the removable membrane surface, thereby facilitating easy removal of deposited sulfur particles (Camiloti et al., 2014; Krayzelova et al., 2015). Additionally, to minimize the issue of excessive substrates oxidation, the application of precise automatic process control for microaeration, such as PID control (Ramos et al., 2014) and ORP-based control with SCADA system (Nghiem et al., 2014) could be used.

5. Future perspectives

In recent years, there has been a growing interest in the applications of microaeration in AD systems to augment organic conversion rate, methane production, H_2S removal, and process stability control. However, to fully understand the underlying mechanisms and pathway of substrate conversion in AD processes associated with microaeration process control, further studies are needed for industrial-scale applications as pointed out:

- Microbial community structure, microbial kinetics, genes expression and metabolic pathway under microaeration condition should be examined using state-of-the-art techniques such as meta'omic analysis and stable isotope labeling. Such information would be helpful in gaining deeper insight into the activity and biochemical pathway of substrate conversion in microaerobic system.
- Long-term effect of microaeration on adaptation of strict anaerobes to free oxygen and ROS needs to be examined.
- Mathematical modeling of microaeration in AD system should be developed and verified with experimental data.
- Oxygen transfer rate and utilization rate under microaerobic condition of different AD reactor configurations and various operating conditions should be established.
- Process control system for oxygen dosing, especially in pilot and industrial-scale AD systems should be developed.

6. Conclusions

Microaeration has been found to have several positive impacts on AD processes ranging from enhancing hydrolysis, methane production, VFAs production to removing hydrogen sulfide and controlling the process stability. Controlling accurate microaeration rate, which stimulates and regulates the activities of facultative and anaerobic bacteria, and methanogens, is critical in achieving the desired objectives. Future works on microbial and molecular analyses along with mathematical modeling and automatic process control systems are needed for the development of microaeration AD system at industrial-scale.

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